

Math 526, F08
Quiz 3

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (12 points) Given

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 5 \\ 0 & 4 & 0 \end{bmatrix}.$$

Find A^{-1} using the Gaussian-Jordan Elimination.

Solution: $[A \ I] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 2 & 4 & 5 & 0 & 1 & 0 \\ 0 & 4 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 2 & 3 & -2 & 1 & 0 \\ 0 & 4 & 0 & 0 & 0 & 1 \end{array} \right] \quad \underline{R_{ow} 2 - 2 \cdot R_{ow} 1}$

$$\xrightarrow{R_3 - 2R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 2 & 3 & -2 & 1 & 0 \\ 0 & 0 & -6 & 4 & -2 & 1 \end{array} \right] \quad \underline{R_{ow} 3 - 2 \cdot R_{ow} 2}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -6 & 4 & -2 & 1 \end{array} \right] \quad \underline{R_{ow} 2 + \frac{1}{2} R_{ow} 3}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & \frac{5}{3} & -\frac{1}{3} & \frac{1}{6} \\ 0 & 2 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -6 & 4 & -2 & 1 \end{array} \right] \quad \underline{R_{ow} 1 + \frac{1}{6} R_{ow} 3}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{5}{3} & -\frac{1}{3} & -\frac{1}{12} \\ 0 & 2 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -6 & 4 & -2 & 1 \end{array} \right] \quad \underline{R_{ow} 1 - \frac{1}{2} R_{ow} 2}$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{5}{3} & -\frac{1}{3} & -\frac{1}{12} \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & -\frac{2}{3} & \frac{1}{3} & -\frac{1}{6} \end{array} \right] = [I \ A^{-1}]$$

$$\Rightarrow A^{-1} = \begin{bmatrix} \frac{5}{3} & -\frac{1}{3} & -\frac{1}{12} \\ 0 & 0 & \frac{1}{4} \\ -\frac{2}{3} & \frac{1}{3} & -\frac{1}{6} \end{bmatrix}$$

Problem 2. (8 points) Find two 2×2 matrices **A** and **B** such that $\mathbf{AB} = \mathbf{0}$ but $\mathbf{A} \neq \mathbf{0}$ and $\mathbf{B} \neq \mathbf{0}$.

Solution: $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \neq \mathbf{0}, \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \neq \mathbf{0}$

$$\mathbf{AB} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$